Knowledge-Based Agent in Artificial intelligence

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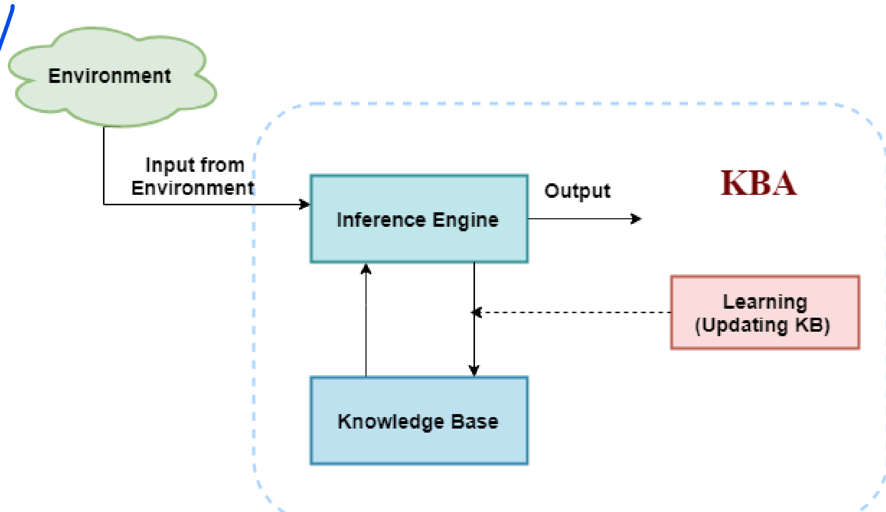
- An intelligent agent needs **knowledge** about the real world for taking decisions and **reasoning** to act efficiently.
- Knowledge-based agents are those agents who have the capability of **maintaining an internal state of knowledge, reason over that knowledge, update their knowledge after observations and take actions. These agents can represent the world with some formal representation and act intelligently**.
- Knowledge-based agents are composed of two main parts:
  - **Knowledge-base and**
  - **Inference system**.

A knowledge-based agent must able to do the following:

- An agent should be able to represent states, actions, etc.
- An agent Should be able to incorporate new percepts
- An agent can update the internal representation of the world
- An agent can deduce the internal representation of the world
- An agent can deduce appropriate actions.

The architecture of knowledge-based agent:



The above diagram is representing a generalized architecture for a knowledge-based agent. The knowledge-based agent (KBA) take input from the environment by perceiving the environment. The input is taken by the inference engine of the agent and which also communicate with KB to decide as per the knowledge store in KB. The learning element of KBA regularly updates the KB by learning new knowledge.

Knowledge base: Knowledge-base is a central component of a knowledge-based agent, it is also known as KB. It is a collection of sentences (here 'sentence' is a technical term and it is not identical to sentence in English). These sentences are expressed in a language which is called a knowledge representation language. The Knowledge-base of KBA stores fact about the world.

Why use a knowledge base?

Knowledge-base is required for updating knowledge for an agent to learn with experiences and take action as per the knowledge.

Inference system

Inference means deriving new sentences from old. Inference system allows us to add a new sentence to the knowledge base. A sentence is a proposition about the world. Inference system applies logical rules to the KB to deduce new information.

Inference system generates new facts so that an agent can update the KB. An inference system works mainly in two rules which are given as:

- **Forward chaining**
- **Backward chaining**

Operations Performed by KBA

Following are three operations which are performed by KBA in order to show the intelligent behavior:

1. **TELL:** This operation tells the knowledge base what it perceives from the environment.
2. **ASK:** This operation asks the knowledge base what action it should perform.
3. **Perform:** It performs the selected action.

A generic knowledge-based agent:

Following is the structure outline of a generic knowledge-based agents program:

```
function KB-AGENT(percept):
persistent: KB, a knowledge base
         t, a counter, initially 0, indicating time
TELL(KB, MAKE-PERCEPT-SENTENCE(percept, t))
Action = ASK(KB, MAKE-ACTION-QUERY(t))
TELL(KB, MAKE-ACTION-SENTENCE(action, t))
 t = t + 1
 return action
```

Book

The knowledge-based agent takes percept as input and returns an action as output. The agent maintains the knowledge base, KB, and it initially has some background knowledge of the real world. It also has a counter to indicate the time for the whole process, and this counter is initialized with zero.

Each time when the function is called, it performs its three operations:

- Firstly it TELLs the KB what it perceives.
- Secondly, it asks KB what action it should take
- Third agent program TELLS the KB that which action was chosen.

The MAKE-PERCEPT-SENTENCE generates a sentence as setting that the agent perceived the given percept at the given time.

The MAKE-ACTION-QUERY generates a sentence to ask which action should be done at the current time.

MAKE-ACTION-SENTENCE generates a sentence which asserts that the chosen action was executed.

Various levels of knowledge-based agent:

A knowledge-based agent can be viewed at different levels which are given below:

1. Knowledge level

Knowledge level is the first level of knowledge-based agent, and in this level, we need to specify what the agent knows, and what the agent goals are. With these specifications, we can fix its behavior. For example, suppose an automated taxi agent needs to go from a station A to station B, and he knows the way from A to B, so this comes at the knowledge level.

2. Logical level:

At this level, we understand that how the knowledge representation of knowledge is stored. At this level, sentences are encoded into different logics. At the logical level, an encoding of knowledge into logical sentences occurs. At the logical level we can expect to the automated taxi agent to reach to the destination B.

3. Implementation level:

This is the physical representation of logic and knowledge. At the implementation level agent perform actions as per logical and knowledge level. At this level, an automated taxi agent actually implement his knowledge and logic so that he can reach to the destination.

Approaches to designing a knowledge-based agent:

There are mainly two approaches to build a knowledge-based agent:

1. **1. Declarative approach:** We can create a knowledge-based agent by initializing with an empty knowledge base and telling the agent all the sentences with which we want to start with. This approach is called Declarative approach.
2. **2. Procedural approach:** In the procedural approach, we directly encode desired behavior as a program code. Which means we just need to write a program that already encodes the desired behavior or agent.

However, in the real world, a successful agent can be built by combining both declarative and procedural approaches, and declarative knowledge can often be compiled into more efficient procedural code.

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Knowledge Representation in AI

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Humans are best at understanding, reasoning, and interpreting knowledge. Human knows things, which is knowledge, and based on their knowledge, they perform various actions in the real world. But how machines do all these things comes under knowledge representation and reasoning. Hence, we can describe Knowledge representation as follows:

Knowledge representation and reasoning (KR, KRR) is the part of Artificial Intelligence which concerned with AI agents' thinking and how thinking contributes to the intelligent behaviour of agents.

It is responsible for representing information about the real world so that a computer can understand and utilise this knowledge to solve complex real-world problems, such as diagnosis a medical condition or communicating with humans in natural language.

It is also a way to describe how we can represent knowledge in artificial intelligence. Knowledge representation is not just storing data in a database, but it also enables an intelligent machine to learn from that knowledge and experience so that it can behave intelligently like a human.

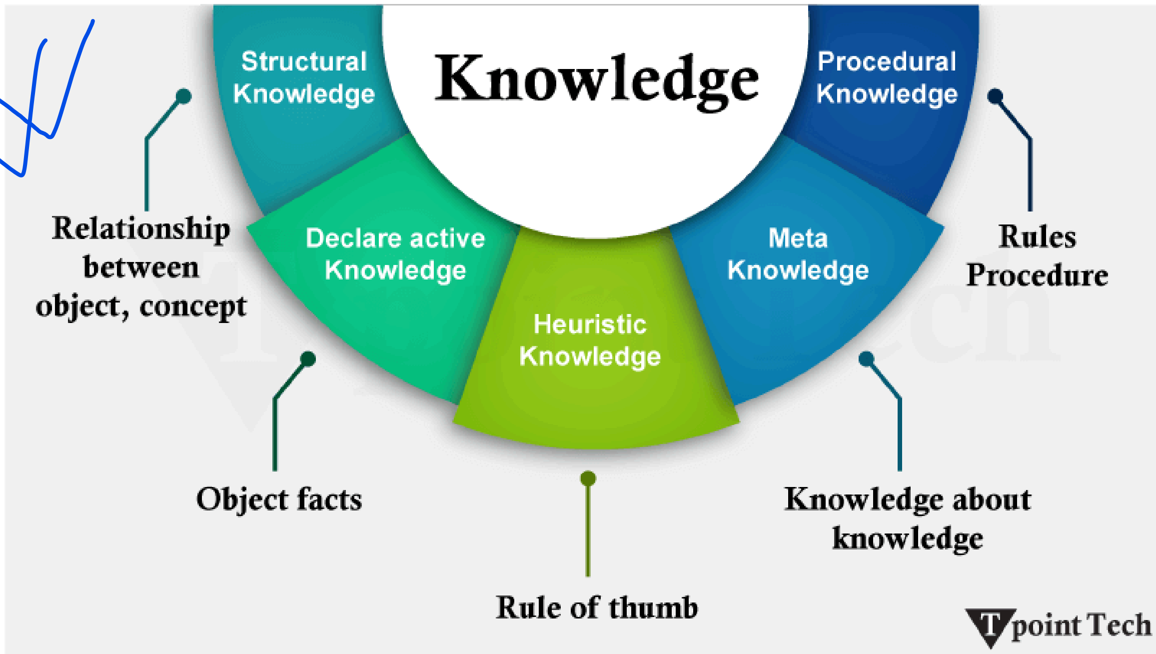
What to Represent?

The following are the kind of knowledge that needs to be represented in AI systems:

- **Object:** All the facts about objects in our world domain. E.g., Guitars contain strings, and trumpets are brass instruments.
- **Events:** Events are the actions that occur in our world.
- **Performance:** It describes behaviour that involves knowledge about how to do things.
- **Meta-knowledge:** It is knowledge about what we know.
- **Facts:** Facts are the truths about the real world and what we represent.
- **Knowledge Base:** The central component of the knowledge-based agents is the knowledge base. It is represented as KB. The Knowledgebase is a group of Sentences (Here, sentences are used as a technical term and not identical with the English language).
- **Knowledge:** Knowledge is awareness or familiarity gained by experiences of facts, data, and situations. The following are the types of knowledge in artificial intelligence:

Types of Knowledge

The following are the various types of knowledge:



1. Declarative Knowledge

- Declarative knowledge is knowing about something.
- It includes concepts, facts, and objects.
- It is also called descriptive knowledge and expressed in declarative sentences.
- It is simpler than procedural language.

2. Procedural Knowledge

- It is also known as imperative knowledge.
- Procedural knowledge is a type of knowledge that is responsible for knowing how to do something.
- It can be directly applied to any task.
- It includes rules, strategies, procedures, agendas, etc.
- Procedural knowledge depends on the task to which it can be applied.

3. Meta-Knowledge

- Knowledge about the other types of knowledge is called Meta-knowledge.

4. Heuristic Knowledge

- Heuristic knowledge represents the knowledge of some experts in a field or subject.
- Heuristic knowledge is rules of thumb based on previous experiences and awareness of approaches, and it is good to work with but not guaranteed.

5. Structural Knowledge

- Structural knowledge is basic knowledge for problem-solving.
- It describes relationships between various concepts such as kind of, part of, and grouping of something.

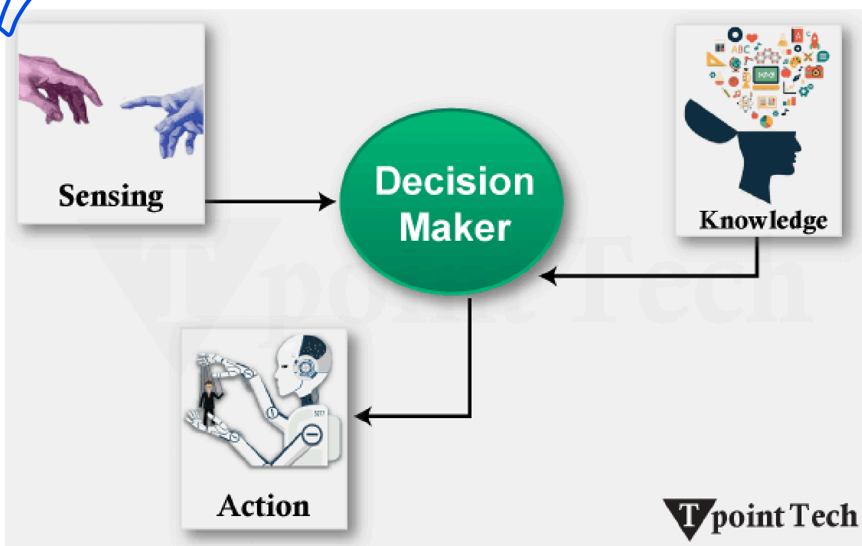
- It describes the relationship that exists between concepts or objects.

The Relation between Knowledge and Intelligence

Knowledge of the real world plays a vital role in intelligence, and the same applies to creating artificial intelligence. Knowledge plays an important role in demonstrating intelligent behaviour in AI agents. An agent is only able to accurately act on some input when he has some knowledge or experience about that input.

Let's suppose that you met a person who is speaking in a language that you don't know; then how would you be able to act on that? The same thing applies to the intelligent behaviour of the agents.

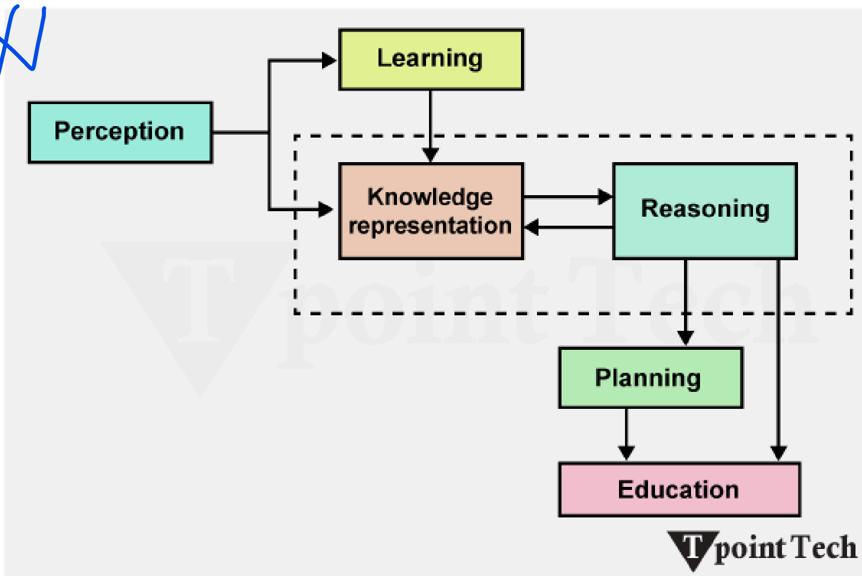
As we can see in the diagram below, there is one decision-maker who acts by sensing the environment and using knowledge. But if the knowledge part is not present, then it cannot display intelligent behaviour.



AI Knowledge Cycle

An Artificial intelligence system has the following components for displaying intelligent behaviour:

- Perception
- Learning
- Knowledge Representation and Reasoning
- Planning
- Execution



The above diagram shows how an AI system can interact with the real world and what components help it to show intelligence. An AI system has a Perception component by which it retrieves information from its environment. It can be visual, audio, or another form of sensory input. The learning component is responsible for learning from data captured by the Perception component.

In the complete cycle, the main components are knowledge representation and Reasoning. These two components are involved in showing the intelligence of machine-like humans. These two components are independent of each other but also coupled together. The planning and execution depend on the analysis of Knowledge representation and reasoning.

Approaches to Knowledge Representation

There are mainly four approaches to knowledge representation, which are given below:

1. Simple Relational Knowledge:

- It is the simplest way of storing facts, which uses the relational method, and each fact about a set of objects is set out systematically in columns.
- This approach of knowledge representation is famous in database systems, where the relationship between different entities is represented.
- This approach has little opportunity for inference.

Example:

The following is a simple relational knowledge representation.

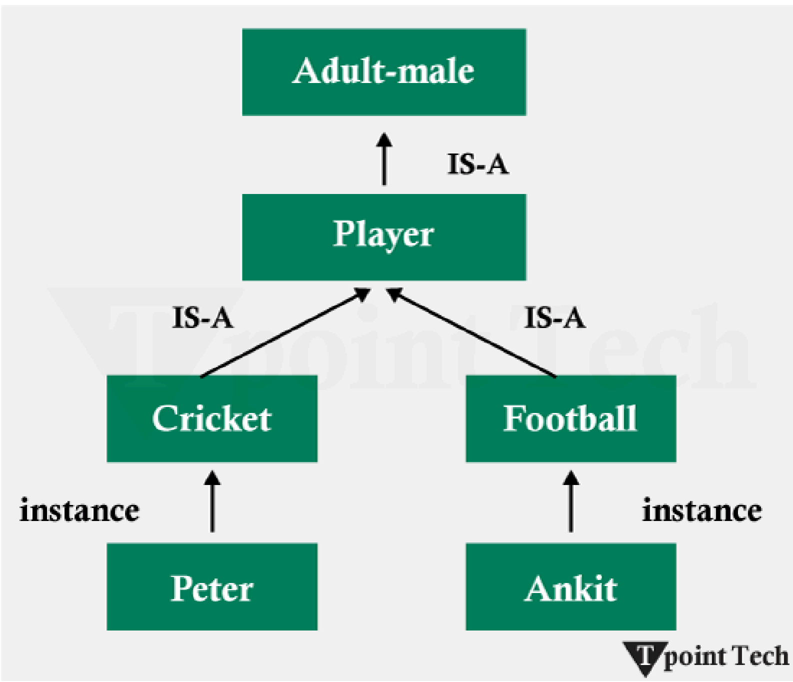
Player	Weight	Age
Player1	65	23
Player2	58	18
Player3	75	24

2. Inheritable Knowledge

- In the inheritable knowledge approach, all data must be stored in a hierarchy of classes.
- All classes should be arranged in a generalised form or a hierarchal manner.

- In this approach, we apply the inheritance property.
- Elements inherit values from other members of a class.
- This approach contains inheritable knowledge, which shows a relation between instance and class, and it is called the instance relation.
- Every individual frame can represent the collection of attributes and their value.
- In this approach, objects and values are represented in Boxed nodes.
- We use Arrows that point from objects to their values.

Example:



3. Inferential Knowledge

- The inferential knowledge approach represents knowledge in the form of formal logic.
- This approach can be used to derive more facts.
- It guaranteed correctness.

Example:

Let's suppose there are two statements:

a. Marcus is a man
b. All men are mortal

Then it can be represented as;

> man(Marcus)
> ∀x = man (x) ----------> mortal (x)s

4. Procedural Knowledge

- The procedural knowledge approach uses small programs and codes that describe how to do specific things and how to proceed.
- In this approach, one important rule is used, which is the If-Then rule.
- With this knowledge, we can use various coding languages such as LISP and Prologue.
- We can easily represent heuristic or domain-specific knowledge using this approach.
- However, it is not necessary that we can represent all cases in this approach.

Requirements for a knowledge Representation system:

A good knowledge representation system must possess the following properties.

1. Representational Accuracy: The KR system should have the ability to represent all kinds of required knowledge.

2. Inferential Adequacy: The KR system should have the ability to manipulate the representational structures to produce new knowledge corresponding to the existing structure.

3. Inferential Efficiency: The ability to direct the inferential knowledge mechanism in the most productive directions by storing appropriate guides.

4. Acquisitional efficiency: The ability to acquire new knowledge easily using automatic methods.

Challenges in Knowledge Representation

Handling Ambiguity and Uncertainty

- **Ambiguity:** In other words, language, symbols, concepts, and other sorts of signifiers can be given more than one meaning, depending on the context. In this case, the word can be any of a financial institution or the side of a river as a bank.
- **Uncertainty:** But, unsurprisingly, incomplete, imprecise, or contradictory information is the fact of the matter. For example, the prediction of stock market trends is uncertainty of economic data, geopolitical factors, as well as human behavior.

They are commonly used to represent and infer under uncertainty with systems that are based on making reasoned guesses using what is known as Bayesian networks and probabilistic graphical models.

Scalability of Representation

- **Volume of Knowledge:** For example, there are plenty of high-dimensional domains involving lots of data to be managed by AI systems, which need to be made available in a timely fashion for use in answering questions (e.g., healthcare or autonomous vehicles).

As we store and retrieve knowledge at a large scale, we can do that using knowledge graphs and distributed storage systems, such as Neo4j, due to which knowledge can be stored and retrieved efficiently.

- **Complexity of Interrelations:** As the knowledge increases, but prior to a point at which computational efficiency is reached, the number of relationships between entities increases.

Clustering, as well as hierarchical representations and modular ontologies, can reduce the complexity in terms of complex relationships without destroying the complexity.

Balancing Expressiveness with Efficiency

Expressiveness:

When you are to be rich and detailed, it is best to know and to get lots of knowledge, but that can be computationally inefficient. For example, ontologies needed to explain such a complex legal framework would prevent the reasoning algorithm.

Approaching the problem, however, they balance the symbolic and the sub-symbolic by giving a problem with symbolist and sub-symbolic representations at the same time. For example, it serves as a framework with which to combine logical frameworks with machine learning models to have high expressiveness without inefficiency.

Efficiency:

Computation is very fast, at the expense of some details that may incur the loss of some information and may lead to suboptimal decisions or reasoning errors. Heuristic-based mechanisms, optimisation algorithms, and, of course, caching mechanisms will facilitate the processing of rich knowledge representations to be efficient.

Dynamic Knowledge Updates and Maintenance

Dynamic Updates:

This becomes the pressing need for the integration of new knowledge without disrupting the existing structure when the domain is changing rapidly, such as weather forecasting or social media analysis. These systems can incrementally learn and online learn (to update their knowledge base without the need to retrain from scratch).

Maintenance:

Such a huge knowledge base that's an ongoing process and incorporates many sources of data makes it difficult to be consistent and accurate. Merging the healthcare databases from several hospitals would include redundant or conflicting information. The risk of quitting is mitigated through regular audits, conflict resolution frameworks, and automatic tools for deduplicating and validating knowledge base quality.

Applications of Knowledge Representation

Problem Solving and Decision Making

- **Structured Problem Analysis:** AI systems can use such frameworks as logical or logical systems and production rules to decompose problems into parts that are easier to manage.
- **Decision Support Systems:** Usually, knowledge graphs and ontologies are used in these systems to assess multiple scenarios, find the best solution, and suggest recommendations.

Robotics and Autonomous Systems

- **Environmental Mapping:** One, that robots use semantic networks to understand spatial relationships to enable pathfinding and navigation, and two, that all commonly implemented application domains that involve navigation can be modelled in semantic networks.
- **Task Automation:** Execution of tasks such as assembling components, cleaning, or delivering goods is made more autonomous by allowing the robot to represent procedural knowledge.
- **Human-Robot Collaboration:** Ontologies allow robots to understand what humans are trying to say and what they intend to do, resulting in natural human-robot interactions in shared spaces.
- **Autonomous Vehicles:** Vehicles can make real-time decisions for safe navigation by being represented as road networks, traffic rules, and environmental context words.

Knowledge-Based Systems in Healthcare and Industry

- **Clinical Decision Support Systems (CDSS):** These systems play a role in helping to diagnose, suggest treatments, and predict patient outcomes by representing medical knowledge in the form of ontologies and rules.
- **Drug Discovery:** The ideal tool for identifying drug targets and studying disease mechanisms is the representation of molecular and genetic data.

- **Predictive Maintenance:** Through knowledge graphs, machines can self-diagnose and predict when a machine will fail, and it saves downtime and cost.
- **Process Optimisation:** Knowledge about production process workflows can be encoded into the system, which can then find and suggest improvements.

Search Engines and Recommender Systems

- **Knowledge Graphs in Search Engines:** Searching the web becomes easy when you represent web content as connected entities (entities are the connected entities). For example, entering "Leonardo da Vinci" results in both web pages as well as facts about his life, his works, and his period.
- **Content-Based Filtering:** By considering the data attributes and modelling similar items, it can help users know their preferences and recommend the preferable items.
- **Collaborative Filtering:** Social and historical data about system users are represented as relationships and used to suggest items that other similar users have liked.

Next Topic Techniques of Knowledge Representation

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Techniques of Knowledge Representation

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Artificial intelligence refers to the process of experiencing intelligence through machines to execute specific functions such as perceiving, understanding, deciding, and deciding. However, this becomes a challenge when it comes to accomplishing this goal because machines need human knowledge to accomplish such tasks. Knowledge representation, which can be defined as the ways and methods that enable the storage and understanding of human knowledge by machines, falls under AI.

There is clearly a significant task with regard to knowledge representation in the context of making it understandable by machines for subsequent use in reasoning and problem-solving. To meet this challenge, several techniques of representing knowledge in artificial intelligence have been formulated, including the rule-based system, semantic network, frame knowledge representation, ontology, and logic-based knowledge representation.

They help organize the information in a manner that such knowledge can be processed and used for the application of varying levels of inference or reasoning.

What is Knowledge Representation?

Knowledge representation was put into practice in an attempt to capture and represent the extent of the relationship between certain concepts, ideas or objects in a way that can elicit inferences or conclusions. In order to do this, four different representation techniques can be employed: logical representation, semantic network representation, frame representation, and production rules.

This makes AI useful in practice in the sense that many intelligent systems are designed using the techniques of knowledge representation in order to reason, understand language, find patterns, learn, and make decisions. For instance, the KRS can be of help in constructing an application that would enable the user to ask questions related to a specific area of interest or create a recommender system to be used to recommend items of interest to the user.

Different Kinds of Knowledge That Need to Be Represented in AI

The knowledge that needs to be represented in AI can be classified as Objects, Events, Performance, Facts, Meta-knowledge or knowledge-base.

Objects

It is a nominal variable defined as things in the external environment that can be viewed in terms of their characteristics or can be GET IT tangible and inert. Some of the objects will be cars, buildings, and people. Various techniques, such as object-oriented programming techniques, represent knowledge in AI.

Events

In wider terms, they refer to activities that occur in the world or actions that happen in the world. Some of the things that are associated with events include driving a car, preparing a meal or going to a concert. Event-based systems are used to represent knowledge in AI, and the use of events does this.

Performance

Performance can also be defined as the manner in which agents or systems act in terms of executing tasks. It consists of the purpose and aims of the task as well as the measures that will be employed to assess productivity. These systems rely on performance as the basis of knowledge in AI.

Facts

Facts mean statements that can either be true or false statements. It is common knowledge that a preposition is a part of speech that involves an adverbial modification of a verb, and it can be confirmed using a fact or as an argument from the conclusion. Some examples of facts include "the sky is blue", "the earth orbits around the sun", and "water boils at 100 degrees Celsius". Invariably, facts are used to model knowledge in AI knowledge-based systems.

Meta-Knowledge

Meta-knowledge refers to knowledge about knowledge. The first subtopic is the structure and organization of knowledge, which is more detailed about the structuring of information and how knowledge institutions are arranged. The meta-knowledge is crucial to AI since it facilitates the evaluation of the quality of knowledge for adequate reasoning to be applied.

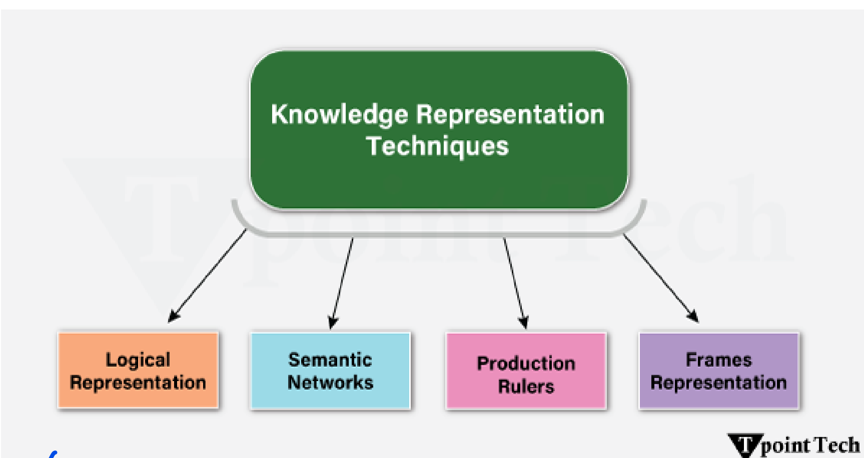
Knowledge-Base

A knowledge base is also referred to as 'artificial knowledge' and can be described as a pool of information in a format that can be accessed and utilized by machines. It is the information that is embedded within an entity and is pertinent to a specific subject area of activity. One of the most used knowledge representations in AI is the use of a knowledge base to represent knowledge in KBS.

Knowledge Representation Techniques

There are four main ways of knowledge representation, which are given as follows:

- Logical Representation
- Semantic Network Representation
- Frame Representation
- Production Rules



Logical Representation

Logical representation is a language with some concrete rules which deal with propositions and have no ambiguity in representation. Logical representation means drawing a conclusion based on various conditions. This representation lays down some important communication rules. It consists of precisely defined syntax and semantics, which support sound inference. Each sentence can be translated into logic using syntax and semantics.

Syntax:

- Syntaxes are the rules that decide how we can construct legal sentences in logic.
- It determines which symbol we can use in knowledge representation.

- How to write those symbols.

Semantics:

- Semantics are the rules by which we can interpret the sentence in the logic.
- Semantics also involves assigning a meaning to each sentence.
- Logical representation can be categorized into mainly two logics:
- Propositional Logics
- Predicate logics

Advantages of logical representation:

- Logical representation enables us to do logical reasoning.
- Logical representation is the basis for the programming languages.

Disadvantages of logical Representation:

- Logical representations have some restrictions and are challenging to work with.
- Logical representation techniques may not be very natural, and inference may not be so efficient.

Semantic Network Representation

Semantic networks are an alternative to predicate logic for knowledge representation. In Semantic networks, we can represent our knowledge in the form of graphical networks. This network consists of nodes representing objects and arcs which describe the relationship between those objects. Semantic networks can categorize objects in different forms and can also link those objects. Semantic networks are easy to understand and can be easily extended.

This representation consists of mainly two types of relations:

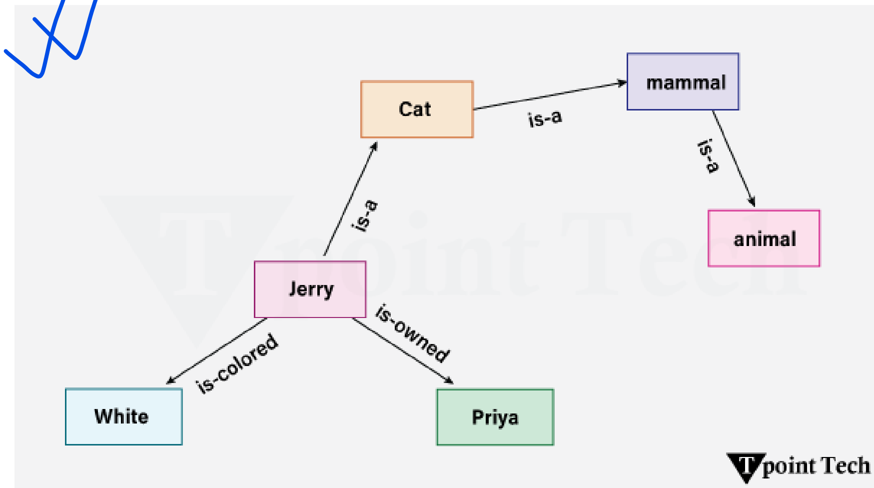
1. IS-A relation (Inheritance)
2. Kind-of-relation

Example:

The following are some statements that we need to represent in the form of nodes and arcs.

Statements:

1. Jerry is a cat.
2. Jerry is a mammal
3. Jerry is owned by Priya.
4. Jerry is brown-coloured.
5. All Mammals are animals.



In the above diagram, we have represented the different types of knowledge in the form of nodes and arcs. Each object is connected with another object by some relation.

Drawbacks in Semantic representation:

- Semantic networks take more computational time at runtime as we need to traverse the complete network tree to answer some questions. It might be possible in the worst-case scenario that after crossing the entire tree, we find that the solution does not exist in this network.
- Semantic networks try to model human-like memory (Which has 1015 neurons and links) to store the information, but in practice, it is not possible to build such a vast semantic network.
- These types of representations are inadequate as they do not have any equivalent quantifier, e.g., for all, for some, none, etc.
- Semantic networks do not have any standard definition for the link names.
- These networks are not intelligent and depend on the creator of the system.

Advantages of Semantic Network:

- Semantic networks are a natural representation of knowledge.
- Semantic networks convey meaning transparently.
- These networks are simple and easily understandable.

Frame Representation

A frame is a record-like structure that consists of a collection of attributes and its values to describe an entity in the world. Frames are the AI data structure that divides knowledge into substructures by representing stereotypical situations. It consists of a collection of slots and slot values. These slots may be of any type and size. Slots have names and values called facets.

Facets: The various aspects of a slot are known as Facets. Facets are features of frames that enable us to put constraints on the frames. Example: IF-NEEDED facts are called when data of any particular slot is needed. A frame may consist of any number of slots, and a slot may include any number of facets, and facets may have any number of values. A frame is also known as slot-filter knowledge representation in artificial intelligence.

Frames are derived from semantic networks and later evolved into our modern-day classes and objects. A single frame is not very useful. The frames system consists of a collection of frames that are connected. In the frame, knowledge about an object or event can be stored together in the knowledge base. The frame is a type of technology that is widely used in various applications, including Natural language processing and machine visions.

Example: 1

Let's take an example of a frame for a book

Slots	Filters
Title	Artificial Intelligence
Genre	Computer Science
Author	Peter Norvig
Edition	Third Edition
Year	1996
Page	1152

Example 2:

Let's suppose we are taking an entity, Peter. Peter is an engineer as a profession, and his age is 25, he lives in the city of London, and the country is England. So, the following is the frame representation for this:

Slots	Filter
Name	Peter
Profession	Doctor
Age	25
Marital status	Single
Weight	78

Advantages of Frame Representation:

- The frame knowledge representation makes the programming easier by grouping the related data.
- The frame representation is comparably flexible and used by many applications in AI.
- It is very easy to add slots for new attributes and relations.
- It is easy to include default data and to search for missing values.
- Frame representation is easy to understand and visualize.

Disadvantages of Frame Representation:

- In the frame system, the inference mechanism is not easily processed.
- The inference mechanism cannot proceed smoothly with a frame representation.
- Frame representation has a much more generalized approach.

Production Rules

Production rules system consists of (condition, action) pairs, which means, "If condition then action". It has mainly three parts:

- The set of production rules
- Working Memory
- The recognize-act-cycle

In production rules, the agent checks for the condition, and if the condition exists, then the production rule fires and a corresponding action is carried out. The condition part of the rule determines which rule may be applied to a problem. The action part carries out the associated problem-solving steps. This complete process is called a recognize-act cycle.

The working memory contains the description of the current state of problem-solving and rules that can write knowledge to the working memory. This knowledge matches and may fire other regulations.

If a new situation (state) is generated, then multiple production rules will be fired together; this is called a conflict set. In this situation, the agent needs to select a rule from these sets, and it is called a conflict resolution.

Example:

- IF (at bus stop AND bus arrives) THEN action (get into the bus)
- IF (on the bus AND paid AND empty seat) THEN action (sit down).
- IF (on bus AND unpaid) THEN action (pay charges).
- IF (bus arrives at destination) THEN action (get down from the bus).

Advantages of Production rule:

- The production rules are expressed in natural language.
- The production rules are highly modular, so we can easily remove, add or modify an individual rule.

Disadvantages of Production rule:

- The production rule system does not exhibit any learning capabilities, as it does not store the result of the problem for future use.
- During the execution of the program, many rules may be active; hence, rule-based production systems are inefficient.

Conclusion

Knowledge representation plays an important role in various intelligent systems where the data has to be stored, processed, and acted upon desirably. Resources like Semantic Networks, Sframes, Rules, and Ontology are other approaches that are used to point toward systematic representation of real-world knowledge. Every one of them has its specific pros and depends on the domain and the degree of the task's complexity. However, rule-based systems are more suitable for logical rather than semantic reasoning and data exchange.

Therefore, ontologies are complementary to rule-based systems. Finally, deciding on an adequate representation scheme is crucial to developing useful, explainable, and provably correct and efficient designs of AI systems. With the development of artificial intelligence, integration of various methods or a combination of advanced approaches is preferred to manage multiple and fluctuating knowledge efficiently.

Next Topic Propositional Logic

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Propositional Logic in Artificial Intelligence

27 May 2025 | 6 min read

Propositional logic is used by artificial intelligence to allow a computer to express propositions concerning a particular subject in formally logical ways. It combines propositions (these are statements that must be either true or false) with logical connectives such as ∧, ∨ and ¬. Many automated logic, knowledge representation and decision-making systems are based on this logic that can be provided.

Given the formal definition of the situations, the so-called facts, which are represented by the so-called propositions, propositional logic provides for an organized approach to the reasoning about situations and, besides, makes the AI system able to conclude new facts based on the available ones.

While less fully formed than predicate logic, due to the limited number of available expressions, first-order logic has a central place in making intelligent agents. It defines a technique of knowledge representation in logical and mathematical form.

Syntax of propositional Logic:

The syntax of propositional logic defines the allowable sentences for the knowledge representation. There are two types of Propositions:

- Atomic Propositions
- Compound propositions

Atomic Proposition: Atomic propositions are simple propositions. It consists of a single proposition symbol. These are the sentences which must be either true or false.

Example 1:

1. 2+2 is 4; it is an atomic proposition as it is a fact.
2. "The Sun is cold" is also a proposition as it is a false fact.

Compound proposition: Compound propositions are constructed by combining simpler or atomic propositions, using parenthesis and logical connectives.

Example 2:

1. "It is raining today, and the street is wet."
2. "Ankit is a doctor, and his clinic is in Mumbai."

Logical Connectives

Logical connectives are used to connect two simpler propositions or represent a sentence logically. We can create compound propositions with the help of logical connectives. There are mainly five connectives, which are given as follows:

- **Negation:** A sentence such as ¬ P is called negation of P. A literal can be either Positive literal or negative literal.
- **Conjunction:** A sentence that has ∧ a connective, such as P ∧ Q, is called a conjunction.

Example: Rohan is intelligent and hardworking. It can be written as,

P= Rohan is intelligent,

Q= Rohan is hardworking. → P∧ Q.

- **Disjunction:** A sentence that has ∨ a connective, such as P ∨, is called disjunction, where P and Q are the propositions.

Example: "Ritika is a doctor or Engineer"

Here P= Ritika is Doctor. Q= Ritika is Doctor, so we can write it as P ∨ Q.

- **Implication:** A sentence such as P → Q is called an implication. Implications are also known as if-then rules. It can be represented as

If it is raining, then the street is wet.

Let P= It is raining, and Q= Street is wet, so it is represented as P → Q

- **Biconditional:** A sentence such as P ⇔ Q is a Biconditional sentence example: If I am breathing, then I am alive

P= I am breathing, Q= I am alive, it can be represented as P ⇔ Q.

Following is the summarized table for Propositional Logic Connectives:

Connective symbols	Word	Technical Term	Example
$\land$	AND	Conjunction	$A \land B$
$\lor$	OR	Disjunction	$A \lor B$
$\rightarrow$	Implies	Implication	$A \rightarrow B$
$\Leftrightarrow$	If and only if	Biconditional	$A \Leftrightarrow B$
$\neg$ or $\sim$	Not	Negation	$\neg A$ or $\neg B$

Truth Table

In propositional logic, we need to know the truth values of propositions in all possible scenarios. We can combine all the possible combinations with logical connectives, and the representation of these combinations in a tabular format is called a Truth table. Following is the truth table for all logical connectives:

For Negation :

P	¬ P
True	False
False	True

For Conjunction:

P	Q	P ∧ Q
True	True	True
True	False	False
False	True	False
False	False	False

For disjunction:

P	Q	P ∨ Q
True	True	True
False	True	True
True	False	True
False	False	False

For Implication:

P	Q	P→Q
True	True	True
True	False	False
False	True	True
False	False	True

For Biconditional :

P	Q	P ⇔ Q
True	True	True
True	False	False
False	True	False
False	False	True

Truth Table with Three Propositions

We can build a proposition composing three propositions: P, Q, and R. This truth table is made up of 8n Tuples as we have taken three proposition symbols.

P	Q	R	¬R	P ∨ Q	P∨Q→¬R
True	True	True	True	True	True
True	True	True	True	True	True
True	True	True	True	True	True
True	True	True	True	True	True
False	False	False	False	False	False
False	False	False	False	False	False
False	False	False	False	False	False
False	False	False	False	False	False

Precedence of connectives:

Just like arithmetic operators, there is a precedence order for propositional connectors or logical operators. This order should be followed while evaluating a propositional problem. Following is the list of the precedence order for operators:

Precedence	Operators
First Precedence	Parenthesis

Second Precedence	Negation
Third Precedence	Conjunction(AND)
Fourth Precedence	Disjunction(OR)
Fifth Precedence	Implication
Six Precedence	Biconditional

Logical Equivalence

Propositional logic is one of the features that have logical equivalence. The definition of logical equivalence is given by saying two propositions are logically equivalent if and only if the columns of a truth table are the same.

Assuming two propositions, A and B, we'll mark it by A⇔B, and this is the logical equivalence. From the below truth table, we see that the columns of ¬A∨ B and B→A are identical, so A is Equivalent to B.

A	B	¬A	¬A∨B	A→B
T	T	F	T	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

Properties of Operators

Commutativity:

P∧ Q= Q ∧ P, or

P ∨ Q = Q ∨ P.

Associativity:

(P ∧ Q) ∧ R= P ∧ (Q ∧ R),

(P ∨ Q) ∨ R= P ∨ (Q ∨ R)

Identity element:

P ∧ True = P,

P ∨ True= True.

Distributive:

P∧ (Q ∨ R) = (P ∧ Q) ∨ (P ∧ R).

P ∨ (Q ∧ R) = (P ∨ Q) ∧ (P ∨ R).

DE Morgan's Law:

¬ (P ∧ Q) = (¬P) ∨ (¬Q)

¬ (P ∨ Q) = (¬ P) ∧ (¬Q).

Double-negation elimination:

¬ (¬P) = P.

Applications of Propositional Logic in AI

- **Knowledge Representation:** Expressed in the propositional logic, knowledge is presented formally in a certain structure. It enables them to store and process information or facts about the world. For instance, in knowledge-based systems, the knowledge base comprises propositions and logical rules.
- **Problem Solving and Planning:** This serves the purpose of allowing AI planners to solve problems and create action sequences given particular goals. For instance, the STRIPS planning system assists propositional logic in the following aspects: it is used in representing preconditions and effects of actions.
- **Decision Making:** If applied properly, it assists in considering top choices and selecting the best approach to be used. Ername logical rules for decision criteria or even truth tables can be implemented in order to evaluate the performance of various choices.
- **Natural Language Processing (NLP):** However, this is also used in NLP applications such as semantic parsing, which involves turning natural language sentences into logical forms. This assists in the meaning post-processing of a sentence and enables one to give reasons for the meaning of a given sentence.

Some Basic Facts about Propositional Logic

- Propositional logic is also called Boolean logic, as it works on 0 and 1.
- In propositional logic, we use symbolic variables to represent the logic, and we can use any symbol to represent a proposition, such as A, B, C, P, Q, R, etc.
- Propositions can be either true or false, but they cannot be both.
- Propositional logic consists of an object, relations or functions, and logical connectives.
- These connectives are also called logical operators.
- Propositions and connectives are the basic elements of propositional logic.
- Connectives can be said as a logical operator which connects two sentences.
- A proposition formula that is always true is called tautology, and it is also called a valid sentence.
- A proposition formula that is always false is called a Contradiction.
- A proposition formula which has both true and false values is called.
- Statements that are questions, commands, or opinions, such as "Where is Rohini?", "How are you?" and "What is your name?" are not propositions.

Limitations of Propositional Logic

Although it has numerous advantages, it also has some drawbacks, and they are as follows:

- **Lack of Expressiveness:** It cannot differentiate scenarios such as 'All humans are mortal'.
- **Scalability:** The Excel table goes up with the number of propositions as the number of rows as rows in the Excel table will increase.
- **Limited Inference:** It only considers and works with true and false propositions and cannot handle the probabilities.
- **No Quantifiers:** Unlike predicate logic, it does not cover the use of quantifiers for all symbols ∝ and their existing symbols ∃.
- **Inability to Handle Uncertainty:** It cannot accommodate probabilities or partial truths to assist, which makes it deficient in uncertain conditions.

- **Lack of Context Awareness:** It removes the meaning or context of statements, which in turn reduces the ability to decipher complex situations.
- We cannot represent relations like ALL, some, or none with propositional logic.

Conclusion

Therefore, propositional logic can be seen as providing essential premises for further improvements in the logical thinking of artificial intelligence. It allows logical relationships to be encoded and represented in such a way that provides a common interface for AI systems to manipulate data and think about it. Even though it is not as explanatory as FOL, PL remains actively employed in rule-based programs, problem-solving paradigms, and planning strategies.

This is particularly important to move on to higher forms of logic, including predicate and probabilistic calculus. In conclusion, it will be pointed out that propositional logic still plays a significant role in the development and work of modern intelligent systems.

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